The Chinese University of Hong Kong, Shenzhen

Course Outline

1. Course identity

A. Course as listed at CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MFE5150
Course title (English)	Financial Data Analysis
Course title (Chinese)	金融数据分析
Units	3
Description (English)	This project-based course provides hands-on training in analyzing real-world financial data using AI-assisted tools. Students will learn to collect, analyze, and extract insights from real financial markets using Python and modern AI code editors. The course emphasizes practical implementation over theoretical derivations, preparing students for quantitative roles in the financial industry.
Description (Chinese)	本项目型课程培训学生使用人工智能辅助工具分析真实金融数据。学生将学习使用 Python 和现代 AI 代码编辑器收集、分析并提取真实金融市场的洞察。课程强调实际操作而非理论推导，旨在为学生准备金融行业的量化岗位。

B. Corresponding course at CUHK

Please give details of the closest corresponding course at CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course at CUHK, please make multiple copies of the block below.

Course code	
Course title (English)	
Course title (Chinese)	
Units	
Description (English)	
Description (Chinese)	

2. Course requisites

Please state prerequisites and co-requisites, in terms of courses at CUHK-Shenzhen or any other requirements.

A. Prerequisites

B. Co-requisites

C. Exclusion

D. Others

3. Learning outcomes

By the end of this course, students will be able to:

- Acquire, clean, and analyze real-world financial data from multiple open-access sources using AI-assisted Python programming.
- Construct and optimize investment portfolios, implement factor models, and develop quantitative trading strategies with proper backtesting methodology.
- Model financial volatility and compute risk metrics such as Value-at-Risk for portfolio risk management.
- Apply machine learning techniques to financial prediction problems while critically assessing model limitations and overfitting risks.
- Use AI code editors effectively as professional tools while maintaining deep understanding of implementation and results.
- Create professional analytical reports and visualizations that communicate findings clearly to various audiences.
- Develop a portfolio of financial analysis projects demonstrating practical, industry-relevant skills to potential employers.
- Think critically about model assumptions, limitations, and the practical challenges of applying quantitative methods in real financial markets.

4. Course syllabus

Please highlight fundamental concepts involved in each topic to help students better understand what is covered in the course.

Portfolio Construction & Optimization, Factor Models & Smart Beta, Volatility Modeling & Risk Management, Time Series for Trading, Machine Learning for Return Prediction, Alternative Data & Modern Applications, Options Pricing & Greeks, AI Agents & RAG for Finance

5. Assessment scheme

Component/ method	% weight
Assignments	10
Project	40
Final	50

Remarks:

6. Grade descriptor

The grade descriptor should align with specific course learning outcomes, not a generic description applicable to all courses.

Type:**General**

Grade	Description
A	Demonstrates the ability to achieve all the learning outcomes in this course and be able to apply the principles in solving problems in novel situations, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study or research. Has the ability to express the synthesis of ideas or application in a clear and cogent manner
A-	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to familiar and standard situations in a manner that is logical and comprehensive. Has the ability to express the knowledge or application with clarity.
B	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.
C	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way.
D	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials Has the ability to state the knowledge or application in simple terms.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

7. Feedback for evaluation

Please specify forms of evaluation to generate feedback on classes from students.

- A course questionnaire answered by every student after finishing all the lectures
- Reflections from teachers of more advanced courses
- Feedback from informal conversation in my office and e-mail correspondence with students after class

8. Readings

Recommended:

1. Forecasting: Principles and Practice, 978-0987507105, R.J. Hyndman, G. Athanasopoulos

Others:

9. Course components

Activity	Hours/week
Lectures	3
Readings & Assignments Readings & Assignments	5
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
1	Introduction. Financial Data Fundamentals & AI Tools
2	Exploratory Data Analysis for Finance
3	Portfolio Construction & Optimization
4	Factor Models & Smart Beta
5	Volatility Modeling & Risk Management
6	Time Series for Trading
7	Time Series for Trading
8	Machine Learning for Return Prediction
9	Alternative Data & Modern Applications
10	Options Pricing & Greeks

11	Advanced Time Series
12	AI Agents & RAG for Finance
13	Final Project Presentations

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections. Example: 150 students for lecture (x 2); 30 students for tutorials (x 10).

Lectures: 100 students in one class
Tutorials: 100 students in one group

12. Academic honesty and plagiarism

A course outline may include subject-specific requirements on plagiarism, in addition to the University established policies.

Students must complete all coursework and assessments independently, unless AI tool use is explicitly permitted. Unauthorized or improper use of AI may constitute academic dishonesty and harm learning. Misuse includes submitting AI-generated work as one's own, using AI to cheat, or employing AI unethically. Always obtain permission before using AI in academic work, and be sure to acknowledge any AI use in submissions.

13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences? Have the details (as in this document) been approved at School or other level in CUHK- Shenzhen?

Submit application

14. Any other information

15. Version date

Version number	1
As of (date)	2025-11-02